Additional file 2: Table of excluded papers

Author	Year	Title	Reason for exclusion
Ranta A, et al.	2014	Utility of a primary care based transient ischemic attack electronic decision support tool: a prospective sequential comparison	Wrong outcome
Yousem DM.	2012	Combating overutilization. Radiology Benefits Managers Versus Order Entry Decision Support	Wrong study design
Langer JM, et al.	2015	Reducing Unnecessary Portable Pelvic Radiographs in Trauma Patients: A Resident-Driven Quality Improvement Initiative	Wrong study design
Ingraham B, et al.	2016	Reduction in High-End Imaging Utilization With Radiology Review and Consultation	Wrong study design
Anandalwar SP, et al.	2016	Eliminating unnecessary routine head CT scanning in neurologically intact mild traumatic brain injury patients: implementation and evaluation of a new protocol	Wrong outcome
Richards SE, et al.	2014	A Clinical Process Change and Educational Intervention to Reduce the Use of Unnecessary Preoperative Tests	Wrong outcome
Schiavoni KH, et al.	2016	How Primary Care Physicians Integrate Price Information into Clinical Decision-Making	Wrong outcome
Jenkins HJ, et al.	2018	Using behavior change theory and preliminary testing to develop an implementation intervention to reduce imaging for low back pain	Wrong study design
Daum P.	2013	Improving the investigation of suspected deep vein thrombosis in the Emergency Department	Wrong study design
Ioos et al.	2011	An integrated approach for prescribing fewer chest x-rays in the ICU	Wrong study design
Hankin RA, Jones SP	2020	The impact of educational interventions on clinicians' knowledge of radiation protection: An integrative review	Wrong outcome
Riggs KR, et al.	2017	Can Appealing to Patient Altruism Reduce Overuse of Health Care Services? An Experimental Survey	Wrong study design
Vest JR, et al.	2015	Image Sharing Technologies and Reduction of Imaging Utilization: A Systematic Review and Meta-analysis	Studies screened for inclusion
Muhiy Addin R, et al.	2020	The Impact of Clinical Decision Support Systems (CDSS) on Physicians: A Scoping Review	Studies screened for inclusion
Deakyne SJ, et al.	2015	Development, Evaluation and Implementation of Chief Complaint Groupings to Activate Data Collection	Wrong outcome
Fenton JJ, et al.	2016	Promoting Patient-Centered Counseling to Reduce Use of Low-Value Diagnostic Tests	Wrong outcome

		A Randomized Clinical Trial	
Gruber J, et al.	2020	The effect of increased cost-sharing on low-value service use	Wrong study design
Jarvik JG, et al.	2020	The Effect of Including Benchmark Prevalence Data of Common Imaging Findings in Spine Image Reports on Health Care Utilization Among Adults Undergoing Spine Imaging. A Stepped-Wedge Randomized Clinical Trial	Wrong outcome
Kullgren JT, et al.	2018	Precommitting to choose wisely about low-value services: a stepped wedge cluster randomised trial	Wrong outcome
Camissa C, et al.	2011	Engaging Physicians in Change: Results of a Safety Net Quality Improvement Program to Reduce Overuse	Wrong outcome
Arditi C, et al.	2017	Computer-generated reminders delivered on paper to healthcare professionals: effects on professional practice and healthcare outcomes	Wrong outcome
Bai L, et al.	2020	A systematic literature review on unnecessary diagnostic testing: The role of ICT use	Wrong outcome
Boutis K, et al.	2015	Cost Consequence Analysis of Implementing the Low Risk Ankle Rule in Emergency Departments	Wrong study design
Bhatia RS, et al.	2014	Educational Intervention to Reduce Inappropriate Transthoracic Echocardiograms: The Need for Sustained Intervention	Wrong study design
Reyes M, et al.	2017	Choosing Wisely Campaign: Report Card and Achievable Benchmarks of Care for Children's Hospitals	Wrong study design
Suman A, et al.	2020	A systematic review of the effectiveness of mass media campaigns for the management of low back pain	Studies screened for inclusion
Psoter KJ, et al.	2014	Effect of an Image-Sharing Network on CT Utilization for Transferred Trauma Patients: A 5-Year Experience at a Level I Trauma Center	Wrong outcome
Ross SE, et al.	2013	Effects of health information exchange adoption on ambulatory testing rates	Wrong outcome
Vest JR, et al.	2014	Health Information Exchange and the Frequency of Repeat Medical Imaging	Wrong outcome
Heubner L, et al.	2019	Using Digital Health to Support Best Practices: Impact of MRI Ordering Guidelines Embedded Within an Electronic Referral Solution	Wrong study design
Fleuren M, et al.	2010	Implementation of a shared care guideline for back pain: effect on unnecessary referrals	Wrong outcome

Chien L, et al.	2019	Reducing low value services in surgical inpatients in Taiwan: Does diagnosis-related group payment work?	Wrong outcome
Gupta A, et al.	2014	Effect of clinical decision support on documented guideline adherence for head CT in emergency department patients with mild traumatic brain injury	Wrong outcome
Kost A, et al.	2015	Clinical Decisions Made in Primary Care Clinics Before and After Choosing Wisely	Wrong outcome